



COMPUTER ORGANISATION (TỔ CHỨC MÁY TÍNH)

Boolean Algebra

Acknowledgement

- The contents of these slides have origin from School of Computing, National University of Singapore.
- We greatly appreciate support from Mr. Aaron Tan Tuck Choy for kindly sharing these materials.

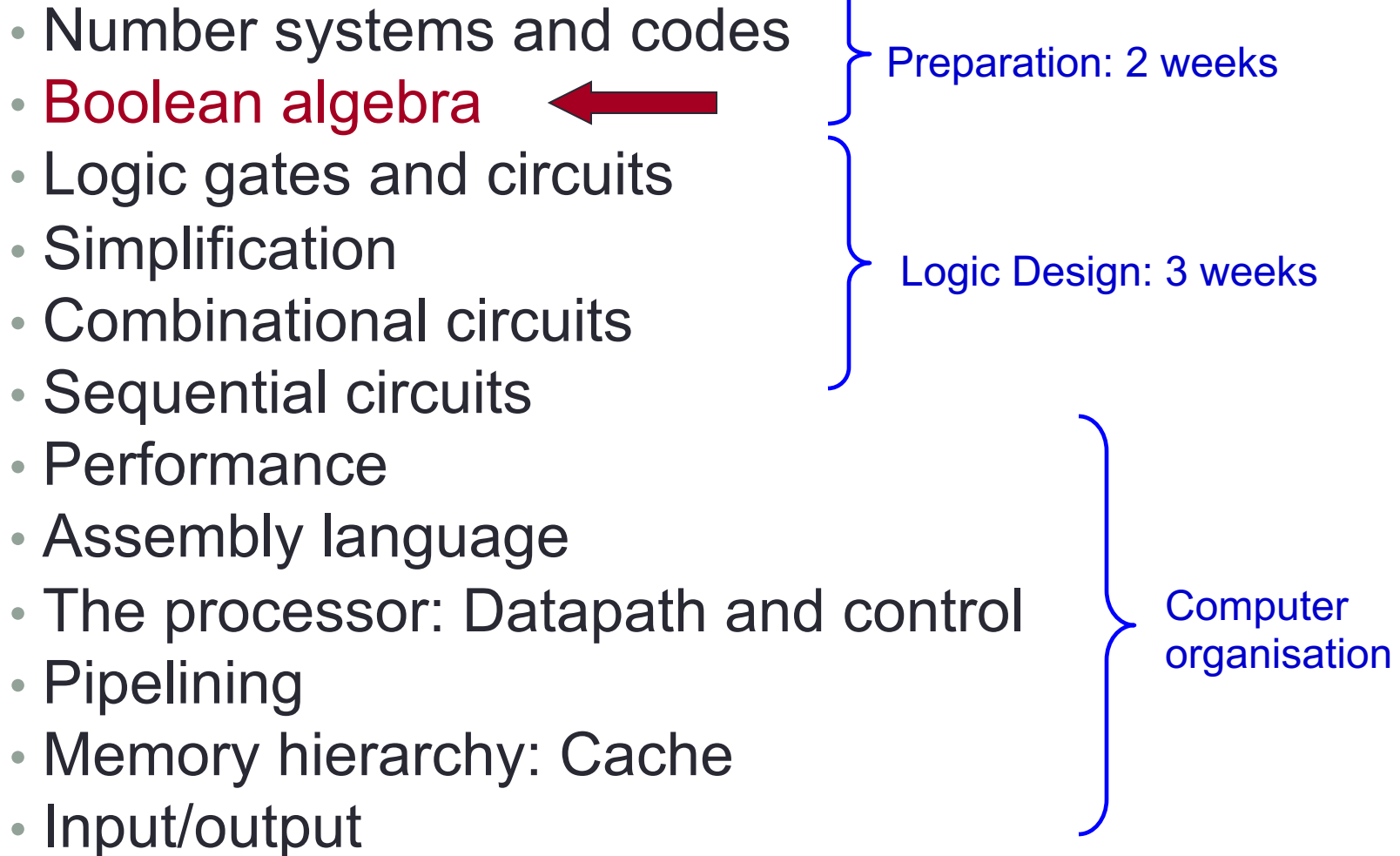
Policies for students

- These contents are only used for students PERSONALLY.
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Recording of modifications

- Minor change in slide 13 (replace new picture)
- Currently, there are no modification on these contents.

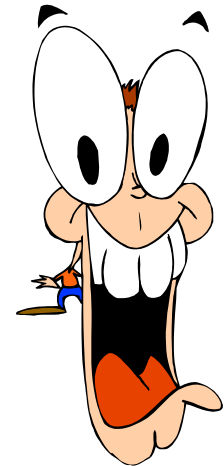
WHERE ARE WE NOW?

- Number systems and codes
 - **Boolean algebra** ←
 - Logic gates and circuits
 - Simplification
 - Combinational circuits
 - Sequential circuits
 - Performance
 - Assembly language
 - The processor: Datapath and control
 - Pipelining
 - Memory hierarchy: Cache
 - Input/output
- Preparation: 2 weeks
- Logic Design: 3 weeks
- Computer organisation
- 



CHECK LIST

- Have you done the *Quick Review Questions* for Chapter 2 Number Systems and Code?
- Have you attempted the *Self-Assessment Exercise #1* on IVLE Assessment?
- Have you clarified your doubts on *IVLE forum*?
- Ready to do a *pop quiz*?



BOOLEAN ALGEBRA

- Boolean Algebra
- Precedence of Operators
- Truth Table
- Duality
- Basic Theorems
- Complement of Functions
- Standard Forms
- Minterms and Maxterms
- Canonical Forms

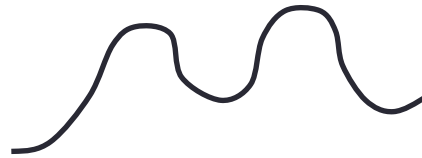
Read up DLD for details!

DIGITAL CIRCUITS (1/2)

- Two voltage levels
 - High, true, 1, asserted
 - Low, false, 0, deasserted



Signals in digital circuit



Signals in analog circuit



A digital watch

DIGITAL CIRCUITS (2/2)

- Advantages of digital circuits over analog circuits
 - More reliable (simpler circuits, less noise-prone)
 - Specified accuracy (determinable)
 - Abstraction can be applied using simple mathematical model – **Boolean Algebra**
 - Ease design, analysis and simplification of digital circuit – **Digital Logic Design**

TYPES OF LOGIC BLOCKS

- **Combinational:** no memory, output depends solely on the input
 - Gates
 - Decoders, multiplexers
 - Adders, multipliers
- **Sequential:** with memory, output depends on both input and current state
 - Counters, registers
 - Memories

BOOLEAN ALGEBRA

- Boolean values:

- True (1)
- False (0)

- Connectives

- Conjunction (AND)
 - $A \cdot B$; $A \wedge B$
- Disjunction (OR)
 - $A + B$; $A \vee B$
- Negation (NOT)
 - $\bar{A}$; A'

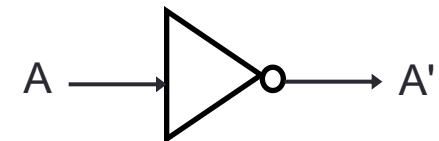
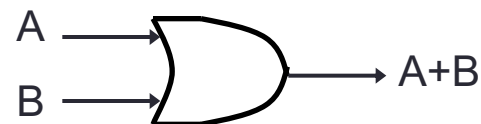
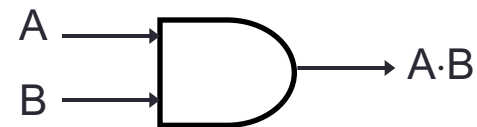
- Truth tables

A	B	$A \cdot B$
0	0	0
0	1	0
1	0	0
1	1	1

A	B	$A + B$
0	0	0
0	1	1
1	0	1
1	1	1

A	A'
0	1
1	0

- Logic gates



AND ($\cdot$)

- Do write the AND operator $\cdot$ instead of omitting it.
 - Example: Write $a \cdot b$ instead of ab
 - Why? Writing ab could mean it is a 2-bit value.



LAWS OF BOOLEAN ALGEBRA

- Identity laws

$$A + 0 = 0 + A = A;$$

$$A \cdot 1 = 1 \cdot A = A$$

- Inverse/complement laws

$$A + A' = 1;$$

$$A \cdot A' = 0$$

- Commutative laws

$$A + B = B + A;$$

$$A \cdot B = B \cdot A$$

- Associative laws

$$A + (B + C) = (A + B) + C;$$

$$A \cdot (B \cdot C) = (A \cdot B) \cdot C$$

- Distributive laws

$$A \cdot (B + C) = (A \cdot B) + (A \cdot C);$$

$$A + (B \cdot C) = (A + B) \cdot (A + C)$$

PRECEDENCE OF OPERATORS

- Precedence from highest to lowest
 - Not
 - And
 - Or
- Examples:
 - $A \cdot B + C = (A \cdot B) + C$
 - $X + Y' = X + (Y')$
 - $P + Q' \cdot R = P + ((Q') \cdot R)$
- Use parenthesis to overwrite precedence. Examples:
 - $A \cdot (B + C)$
 - $(P + Q)' \cdot R$

TRUTH TABLE

- Provide a listing of every possible combination of inputs and its corresponding outputs.
 - Inputs are usually listed in binary sequence.
- **Example**
 - Truth table with 3 inputs and 2 outputs

x	y	z	$y + z$	$x \cdot (y + z)$
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	1	0
1	0	0	0	0
1	0	1	1	1
1	1	0	1	1
1	1	1	1	1

PROOF USING TRUTH TABLE

- **Prove:** $x \cdot (y + z) = (x \cdot y) + (x \cdot z)$
 - Construct truth table for LHS and RHS

x	y	z	$y + z$	$x \cdot (y + z)$	$x \cdot y$	$x \cdot z$	$(x \cdot y) + (x \cdot z)$
0	0	0					
0	0	1					
0	1	0					
0	1	1					
1	0	0					
1	0	1					
1	1	0					
1	1	1					

- Check that column for LHS = column for RHS

DUALITY

- If the AND/OR operators and identity elements 0/1 in a Boolean equation are interchanged, it remains valid
- Example:
 - The dual equation of $a+(b \cdot c)=(a+b) \cdot (a+c)$ is $a \cdot (b+c)=(a \cdot b)+(a \cdot c)$
- Duality gives free theorems – “two for the price of one”. You prove one theorem and the other comes for free!
- Examples:
 - If $(x+y+z)' = x' \cdot y' \cdot z'$ is valid, then its dual is also valid:
 $(x \cdot y \cdot z)' = x' + y' + z'$
 - If $x+1 = 1$ is valid, then its dual is also valid:
 $x \cdot 0 = 0$

BASIC THEOREMS (1/2)

1. Idempotency

$$X + X = X ;$$

$$X \cdot X = X$$

2. Zero and One elements

$$X + 1 = 1 ;$$

$$X \cdot 0 = 0$$

3. Involution

$$(X')' = X$$

4. Absorption

$$X + X \cdot Y = X ;$$

$$X \cdot (X + Y) = X$$

5. Absorption (variant)

$$X + X' \cdot Y = X + Y ;$$

$$X \cdot (X' + Y) = X \cdot Y$$

BASIC THEOREMS (2/2)

6. DeMorgan's

$$(X + Y)' = X' \cdot Y' ;$$

$$(X \cdot Y)' = X' + Y'$$

DeMorgan's Theorem can be generalised to more than two variables, example: $(A + B + \dots + Z)' = A' \cdot B' \cdot \dots \cdot Z'$

7. Consensus

$$X \cdot Y + X' \cdot Z + Y \cdot Z = X \cdot Y + X' \cdot Z$$

$$(X+Y) \cdot (X'+Z) \cdot (Y+Z) = (X+Y) \cdot (X'+Z)$$

PROVING A THEOREM

- Theorems can be proved using truth table, or by algebraic manipulation using other theorems/laws.
- Example: Prove absorption theorem $X + X \cdot Y = X$
$$\begin{aligned} X + X \cdot Y &= X \cdot 1 + X \cdot Y \text{ (by identity)} = X \cdot (1 + Y) \text{ (by distributivity)} \\ &= X \cdot (Y + 1) \text{ (by commutativity)} = X \cdot 1 \text{ (by one element)} \\ &= X \text{ (by identity)} \end{aligned}$$
- By duality, we have also proved $X \cdot (X + Y) = X$

BOOLEAN FUNCTIONS

- Examples of Boolean functions (logic equations):

$$F1(x,y,z) = x \cdot y \cdot z'$$

$$F2(x,y,z) = x + y' \cdot z$$

$$F3(x,y,z) = x' \cdot y' \cdot z + x' \cdot y \cdot z + x \cdot y'$$

$$F4(x,y,z) = x \cdot y' + x' \cdot z$$

x	y	z	F1	F2	F3	F4
0	0	0	0			
0	0	1	0			
0	1	0	0			
0	1	1	0			
1	0	0	0			
1	0	1	0			
1	1	0	1			
1	1	1	0			

COMPLEMENT

- Given a Boolean function F , the **complement** of F , denoted as F' , is obtained by interchanging 1 with 0 in the function's output values.
- Example: $F1 = x \cdot y \cdot z'$
- What is $F1'$?

x	y	z	F1	F1'
0	0	0	0	
0	0	1	0	
0	1	0	0	
0	1	1	0	
1	0	0	0	
1	0	1	0	
1	1	0	1	
1	1	1	0	

STANDARD FORMS (1/2)

- Certain types of Boolean expressions lead to circuits that are desirable from implementation viewpoint.
- Two standard forms:
 - Sum-of-Products
 - Product-of-Sums
- Literals
 - A Boolean variable on its own or in its complemented form
 - Examples: x , x' , y , y'
- Product term
 - A single literal or a logical product (AND) of several literals
 - Examples: x , $x \cdot y \cdot z'$, $A' \cdot B$, $A \cdot B$, $d \cdot g' \cdot v \cdot w$

STANDARD FORMS (2/2)

- **Sum term**

- A single literal or a logical sum (OR) of several literals
- Examples: x , $x+y+z'$, $A'+B$, $A+B$, $c+d+h'+j$

- **Sum-of-Products (SOP) expression**

- A product term or a logical sum (OR) of several product terms
- Examples: x , $x + y \cdot z'$, $x \cdot y' + x' \cdot y \cdot z$, $A \cdot B + A' \cdot B'$,
 $A + B' \cdot C + A \cdot C' + C \cdot D$

- **Product-of-Sums (POS) expression**

- A sum term or a logical product (AND) of several sum terms
- Examples: x , $x \cdot (y+z')$, $(x+y') \cdot (x'+y+z)$,
 $(A+B) \cdot (A'+B')$, $(A+B+C) \cdot D' \cdot (B'+D+E')$

- **Every Boolean expression can be expressed in SOP or POS.**

DO IT YOURSELF

SOP expr: A product term or a logical sum (OR) of several product terms.

POS expr: A sum term or a logical product (AND) of several sum terms.

- Put the right ticks in the following table.

<i>Expression</i>	<i>SOP?</i>	<i>POS?</i>
$X' \cdot Y + X \cdot Y' + X \cdot Y \cdot Z$		
$(X + Y') \cdot (X' + Y) \cdot (X' + Z')$		
$X' + Y + Z$		
$X \cdot (W' + Y \cdot Z)$		
$X \cdot Y \cdot Z'$		
$W \cdot X' \cdot Y + V \cdot (X \cdot Z + W')$		

MINTERMS & MAXTERMS (1/2)

- A **minterm** of n variables is a product term that contains n literals from all the variables.
 - Example: On 2 variables x and y , the minterms are:
 $x' \cdot y'$, $x' \cdot y$, $x \cdot y'$ and $x \cdot y$
- A **maxterm** of n variables is a sum term that contains n literals from all the variables.
 - Example: On 2 variables x and y , the maxterms are:
 $x' + y'$, $x' + y$, $x + y'$ and $x + y$
- In general, with n variables we have 2^n minterms and 2^n maxterms.

MINTERMS & MAXTERMS (2/2)

- The minterms and maxterms on 2 variables are denoted by **m0 to m3** and **M0 to M3** respectively.

x	y	Minterms		Maxterms	
		Term	Notation	Term	Notation
0	0	$x' \cdot y'$	m0	$x+y$	M0
0	1	$x' \cdot y$	m1	$x+y'$	M1
1	0	$x \cdot y'$	m2	$x'+y$	M2
1	1	$x \cdot y$	m3	$x'+y'$	M3

- Each minterm is the complement of the corresponding maxterm
 - Example: $m_2 = x \cdot y'$
 $m_2' = (x \cdot y')' = x' + (y')' = x' + y = M_2$

CANONICAL FORMS

- Canonical/normal form: a unique form of representation.
 - Sum-of-minterms = Canonical sum-of-products
 - Product-of-maxterms = Canonical product-of-sums

SUM-OF-MINTERMS

- Given a truth table, example:
- Obtain **sum-of-minterms** expression by gathering the minterms of the function (where output is 1).

$$F1 = x \cdot y \cdot z' = m6$$

$$F2 =$$

$$F3 =$$

x	y	z	F1	F2	F3
0	0	0	0	0	0
0	0	1	0	1	1
0	1	0	0	0	0
0	1	1	0	0	1
1	0	0	0	1	1
1	0	1	0	1	1
1	1	0	1	1	0
1	1	1	0	1	0

PRODUCT-OF-MAXTERMS

- Given a truth table, example:
- Obtain **product-of-maxterms** expression by gathering the maxterms of the function (where output is 0).

x	y	z	F1	F2	F3
0	0	0	0	0	0
0	0	1	0	1	1
0	1	0	0	0	0
0	1	1	0	0	1
1	0	0	0	1	1
1	0	1	0	1	1
1	1	0	1	1	0
1	1	1	0	1	0

$$\begin{aligned}
 F2 &= (x+y+z) \cdot (x+y'+z) \cdot (x+y'+z') \\
 &= M0 \cdot M2 \cdot M3 = \prod M(0,2,3)
 \end{aligned}$$

$$F3 =$$

CONVERSION

- We can convert between sum-of-minterms and product-of-maxterms easily
- Example: $F2 = \Sigma m(1,4,5,6,7) = \Pi M(0,2,3)$
- Why? See $F2'$ in truth table.

■ $F2' = m0 + m2 + m3$
 Therefore,
 $F2 = (m0 + m2 + m3)'$
 $= m0' \cdot m2' \cdot m3'$ (by DeMorgan's)
 $= M0 \cdot M2 \cdot M3$ ($mx' = Mx$)

x	y	z	F2	F2'
0	0	0	0	1
0	0	1	1	0
0	1	0	0	1
0	1	1	0	1
1	0	0	1	0
1	0	1	1	0
1	1	0	1	0
1	1	1	1	0

READING ASSIGNMENT

- **Conversion of Standard Forms**
 - Read up DLD section 3.4, pg 57 – 58.



Q&A